

02443 Stochastic Simulation

Day 6: Bootstrapping

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The Bootstrap Method

In simulation studies and statistical analyses, we often estimate parameters from data.

Examples include:

- the mean,
- the median,
- quantiles,
- probabilities,
- regression coefficients.

In addition to the estimate itself, we are often interested in its uncertainty.

How can we estimate the variance of an estimator when no analytical expression is available?

The Bootstrap Method

Bootstrapping is a non-parametric technique for estimating:

- variances,
- standard errors,
- confidence intervals,

of statistical estimators.

Basic idea

Suppose we observe a sample

$$x_1, x_2, \dots, x_n.$$

The true underlying distribution is unknown.

Instead of sampling from the true distribution, we sample from the empirical distribution constructed from the observed data.

Observed data $\longrightarrow$ Empirical distribution $\longrightarrow$ New samples

A simple situation

Consider a sample $\{X_i\}_{i=1}^n$ with n observations.

If we wish to estimate the mean of the underlying distribution, a natural estimator is the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Assuming the observations are independent and identically distributed,

$$\mathbb{V}[\bar{X}] = \frac{\mathbb{V}[X_i]}{n}.$$

Since $\mathbb{V}[X_i]$ can be estimated using the sample variance, estimating the variance of the sample mean is straightforward.

A less simple situation

Suppose instead that we wish to estimate the median of the underlying distribution.

A natural estimator is the sample median. The sample median is often preferred because it is more robust to outliers than the sample mean.

However, a natural question arises:

What is the variance of the sample median?

Unlike the sample mean, there is generally no simple formula for the variance of the sample median.

This is exactly the type of problem that motivates the bootstrap method.

Estimating the variance of the sample median

Suppose we wish to estimate the variance of the sample median.

If the true distribution were known

We could estimate this variance using simulation:

- 1 Simulate many datasets of size n from the true distribution.
- 2 Compute the sample median for each dataset.
- 3 Calculate the variance among the simulated medians.

The resulting variance would provide an estimate of $\mathbb{V}[\hat{m}]$, where $\hat{m}$ denotes the sample median.

Unfortunately, the true distribution is usually unknown.

The bootstrap principle

Although the true distribution is unknown, we do have the observed data

$$x_1, x_2, \dots, x_n.$$

From these observations, we can construct the empirical distribution.

The bootstrap principle is:

Replace the unknown distribution by the empirical distribution.

We then proceed exactly as before:

- 1 Simulate many datasets from the empirical distribution.
- 2 Compute the estimator of interest for each dataset.
- 3 Estimate the variance from the resulting collection of estimates.

The empirical distribution acts as a surrogate for the unknown true distribution.

The empirical distribution

The empirical distribution assigns probability mass $1/n$ to each observed data point. Consequently, observations appearing multiple times receive proportionally more probability mass.

Empirical distribution function

The empirical distribution function is defined as

$$F_e(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{x_i \leq x\}.$$

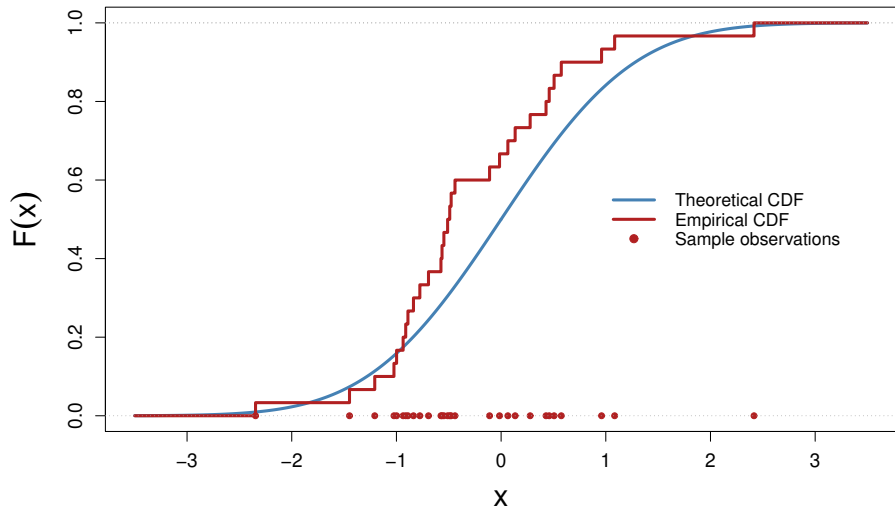
For a given value of x :

- Count the number of observations satisfying $x_i \leq x$.
- Divide by n .

Thus, $F_e(x)$ is simply the proportion of observations less than or equal to x .

The empirical distribution

Theoretical and Empirical CDF



The empirical distribution - An exercise

Let X be a sample with n independent observations from the distribution F .

Expectation

Derive the expectation

$$\mathbb{E}[F_e(x)].$$

Variance

Derive the variance

$$\mathbb{V}[F_e(x)].$$

The algorithm

We have the data $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and want to estimate θ . We assume that the observations are independent and identically distributed. We use the original sample to produce the estimate $\hat{\theta}$, but use bootstrapping to estimate the uncertainty of the estimate.

- 1 Generate k bootstrap samples, each with n observations sampled with replacement from the original dataset.
- 2 Calculate a bootstrap replicate $\hat{\theta}_i^*$ for each bootstrap sample.
- 3 Calculate the sample mean of the k bootstrap replicates

$$\bar{\theta}^* = \frac{1}{k} \sum_{i=1}^k \hat{\theta}_i^*$$

and estimate of the variance of $\hat{\theta}$ using the sample variance

$$\hat{V}[\hat{\theta}] = \frac{1}{k-1} \sum_{i=1}^k (\hat{\theta}_i^* - \bar{\theta}^*)^2.$$

Advantages of bootstrapping

The bootstrap method has several attractive properties:

- Does not require a parametric model for the underlying distribution.
- Easy to implement using simulation.
- Applies to a wide range of estimators.
- Particularly useful when analytical variance formulas are unavailable.
- Can be extended to confidence intervals and bias estimation.

The bootstrap replaces mathematical analysis by simulation whenever the latter is easier.

Exercise 8

Part 1

Solve exercise 13 in Chapter 8 of Ross (p.152).

Part 2

Solve exercise 15 in Chapter 8 of Ross (p.152).

Part 3

Write a program that takes a sample as input and returns the sample median together with a bootstrap estimate of the variance of the median, based on $k = 100$ bootstrap replicates. Then run the program with a sample of $n = 200$ Pareto($\beta = 1, k = 1.05$) distributed observations.

- 1 Compute the sample mean and sample median.
- 2 Estimate the variance of the sample mean using the bootstrap method.
- 3 Estimate the variance of the sample median using the bootstrap method.
- 4 Compare the precision of the two estimators.
- 5 Explain the observed differences in terms of the underlying distribution.